

Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the `plot()` function and add on packages like **ggplot2**.

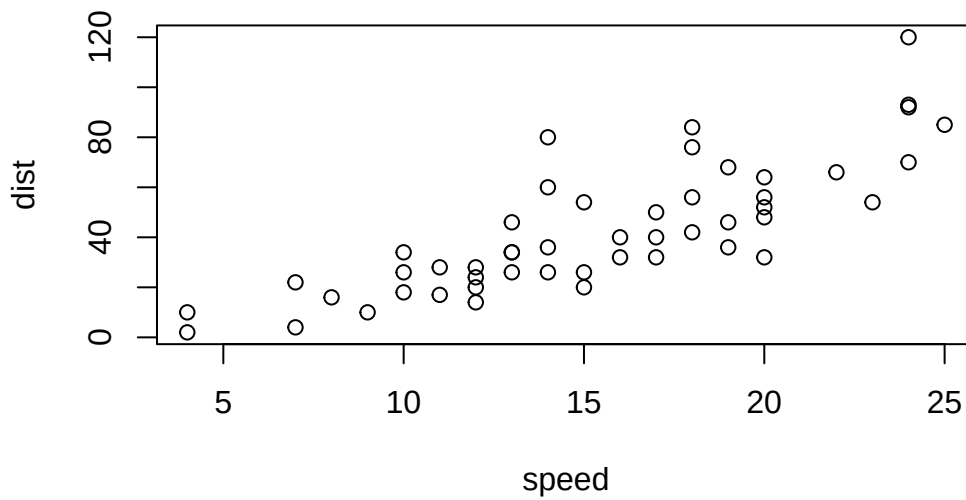
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use ggplot2 I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

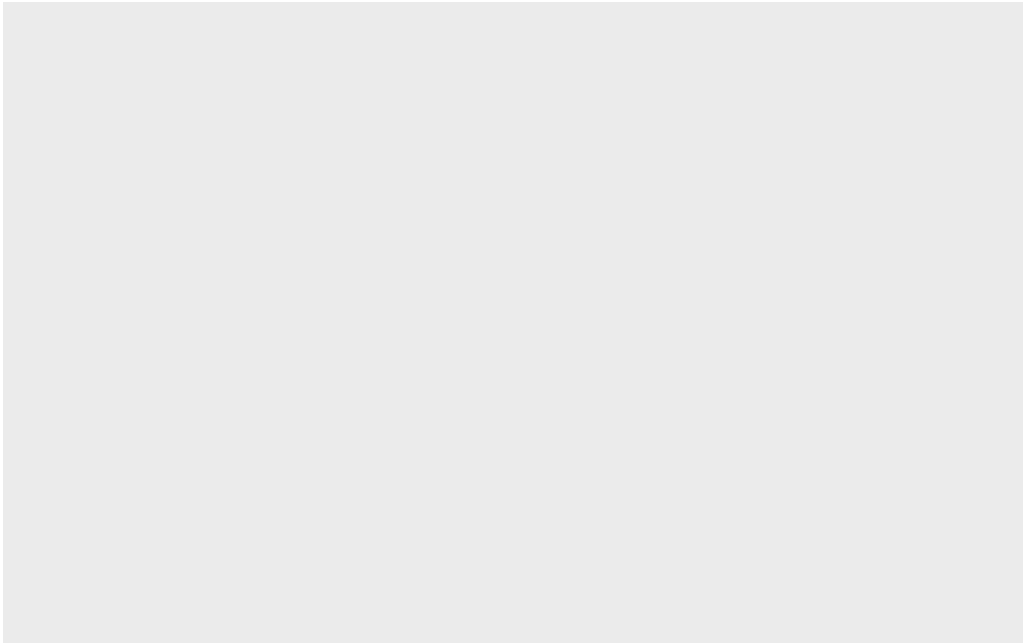
N.B. We never run `install.packages()` in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our quarto report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called `ggplot()`

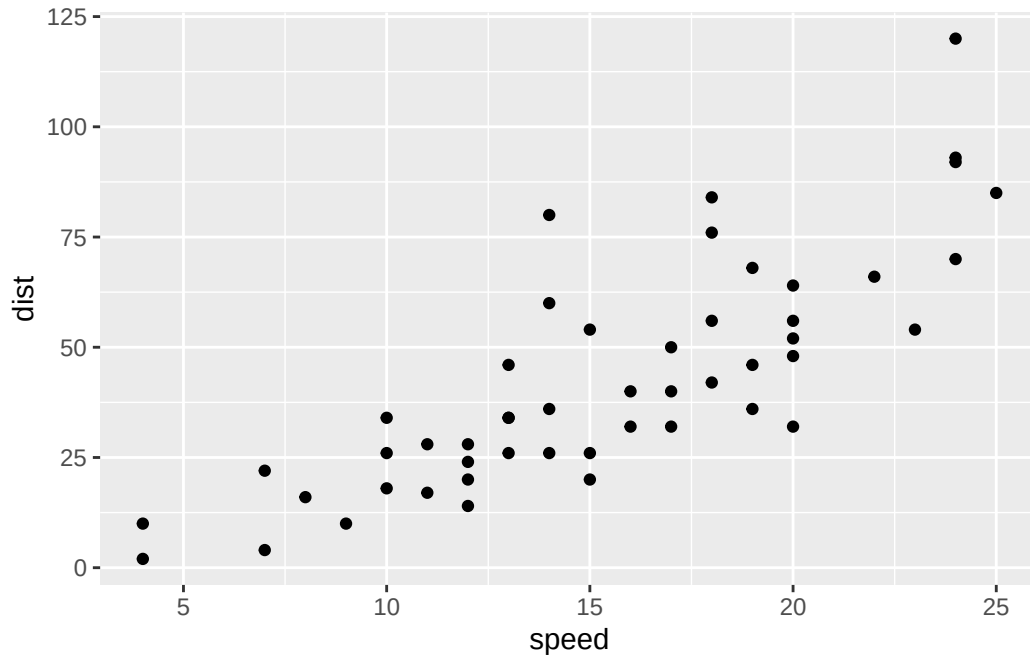
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot),
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

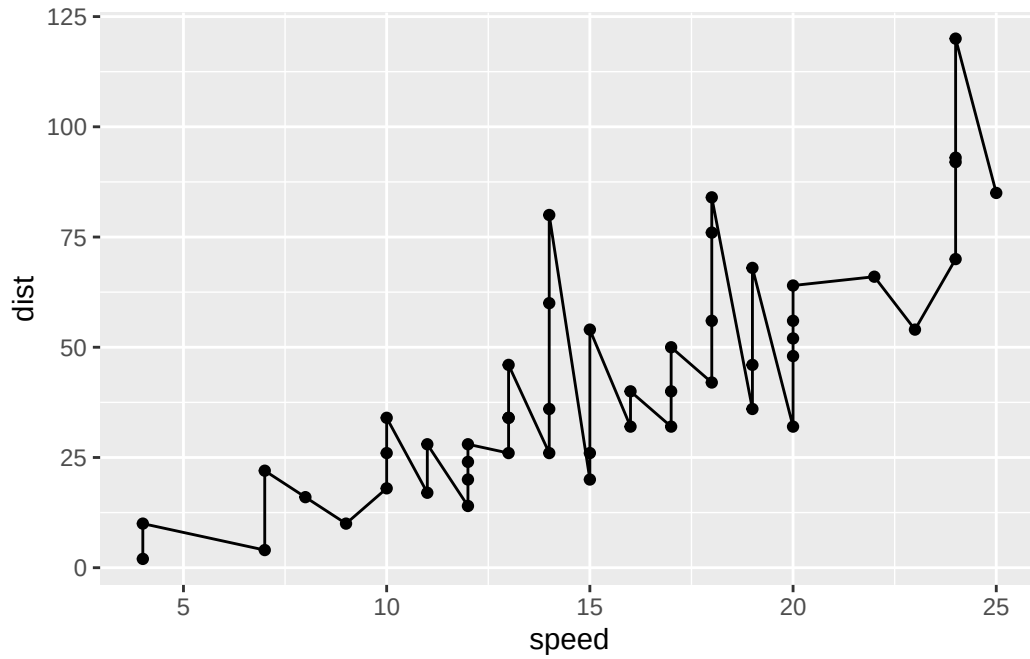
```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point()
```



Add some custom features

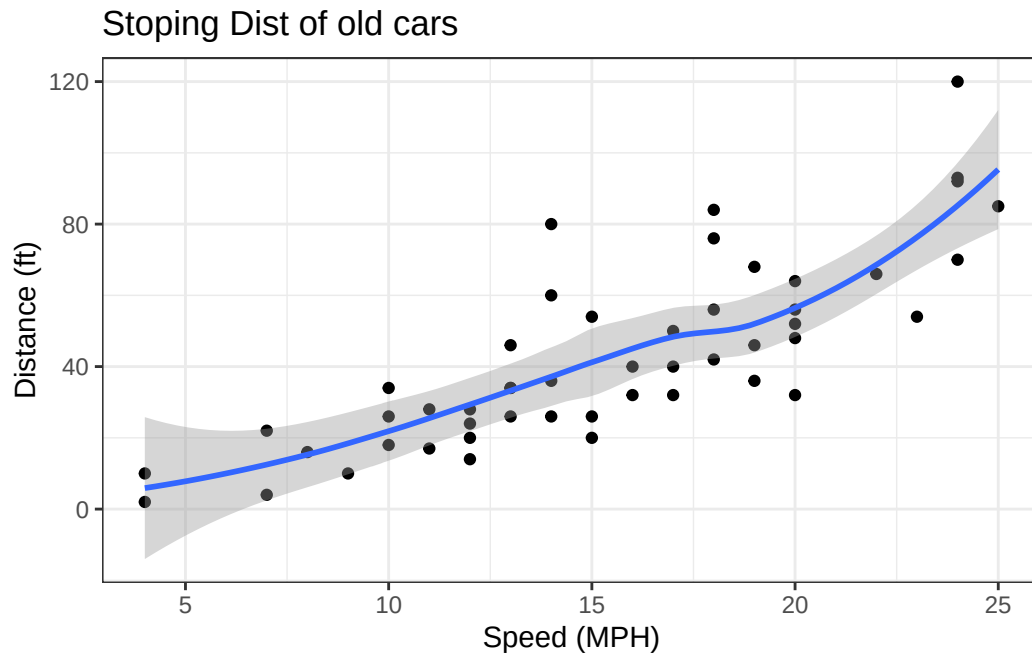
Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point() +  
  geom_line()
```



```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point() +  
  geom_smooth() +  
  theme_bw() +  
  labs(title = "Stopping Dist of old cars",  
        x = "Speed (MPH)",  
        y = "Distance (ft)")
```

`geom_smooth()` using method = 'loess' and formula = 'y ~ x'

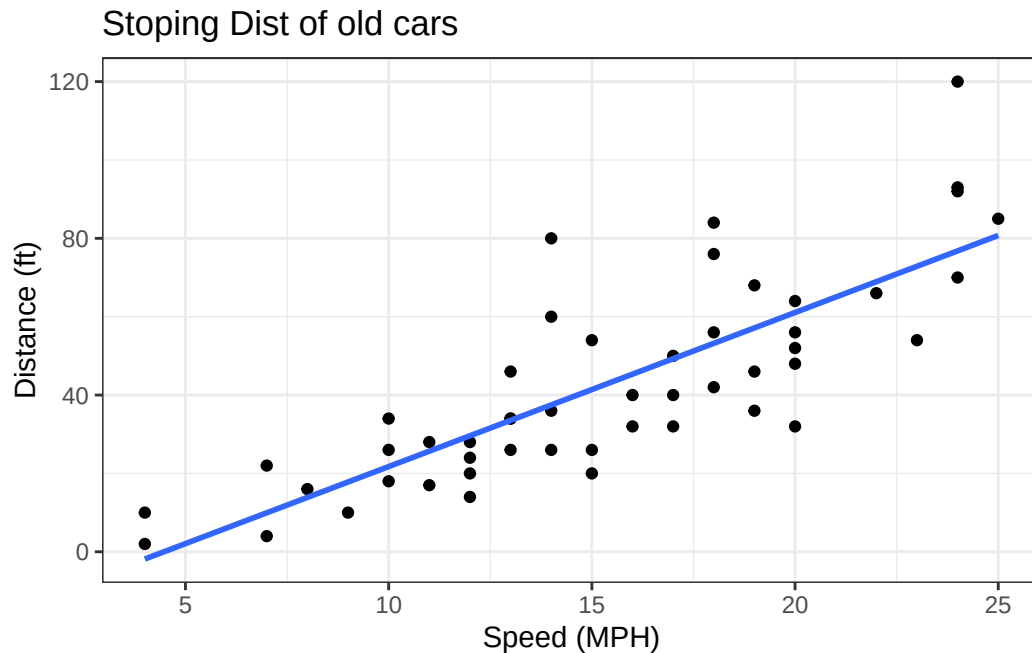


Q. Can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the “gray” error region.

Gene Expression Figure

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw() +
  labs(title = "Stopping Dist of old cars",
       x = "Speed (MPH)",
       y = "Distance (ft)")
```

``geom_smooth()`` using formula = `'y ~ x'`



```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

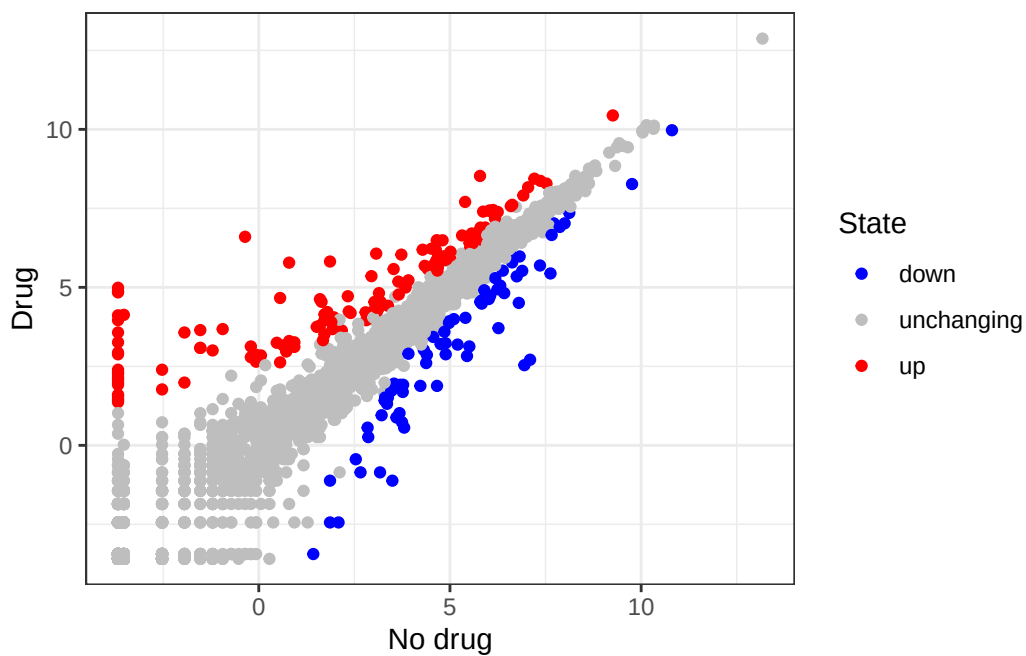
A useful new function in this context is the `table()` function:

```
table(genes$State)
```

down	unchanging	up
72	4997	127

My first plot attempt

```
ggplot(genes) +
  aes(Condition1, Condition2, col=State) +
  geom_point() +
  scale_color_manual(values = c("blue","gray","red")) +
  theme_bw() +
  labs(x= "No drug", y = "Drug")
```



Going further

Here we read the famous gapminder dataset.

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```


	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different “country” entries are in this dataset?

```
length (table(gapminder$country))
```

```
[1] 142
```

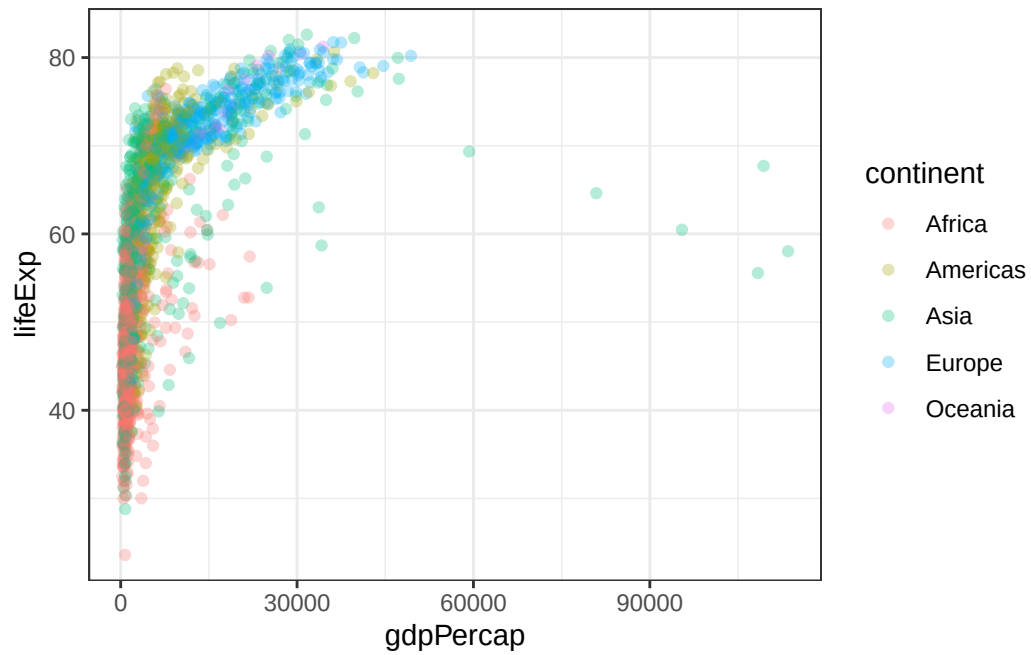
```
length (unique (gapminder$country))
```

```
[1] 142
```

Let’s make our first plot of the entire dataset:

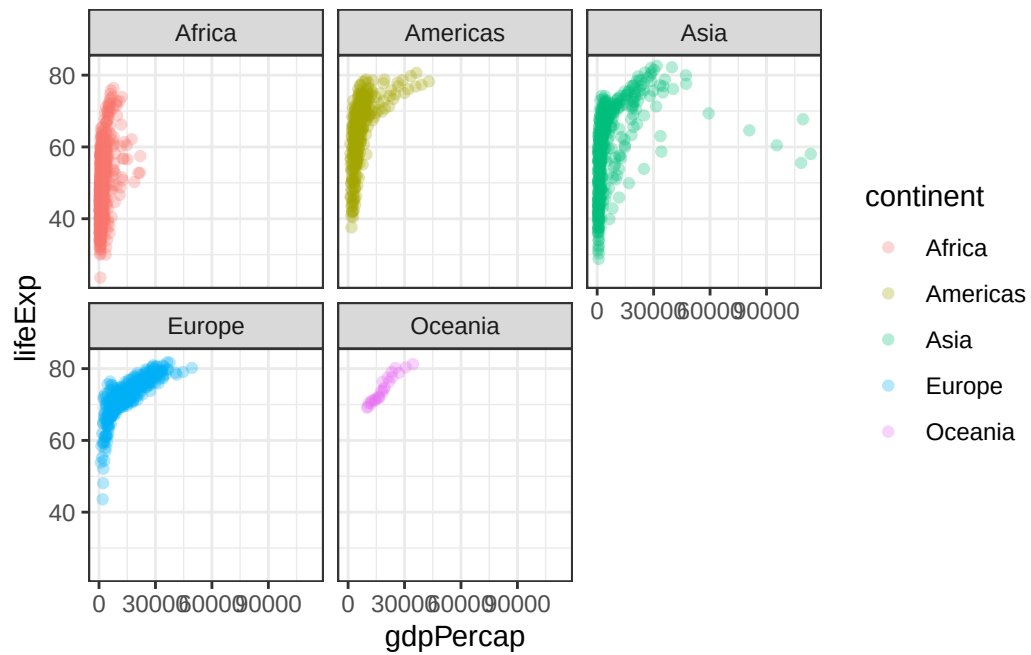
```
p <- ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha=0.3) +
  theme_bw()
```

```
p
```



I can add more layers to p

```
p +  
  facet_wrap(~continent)
```



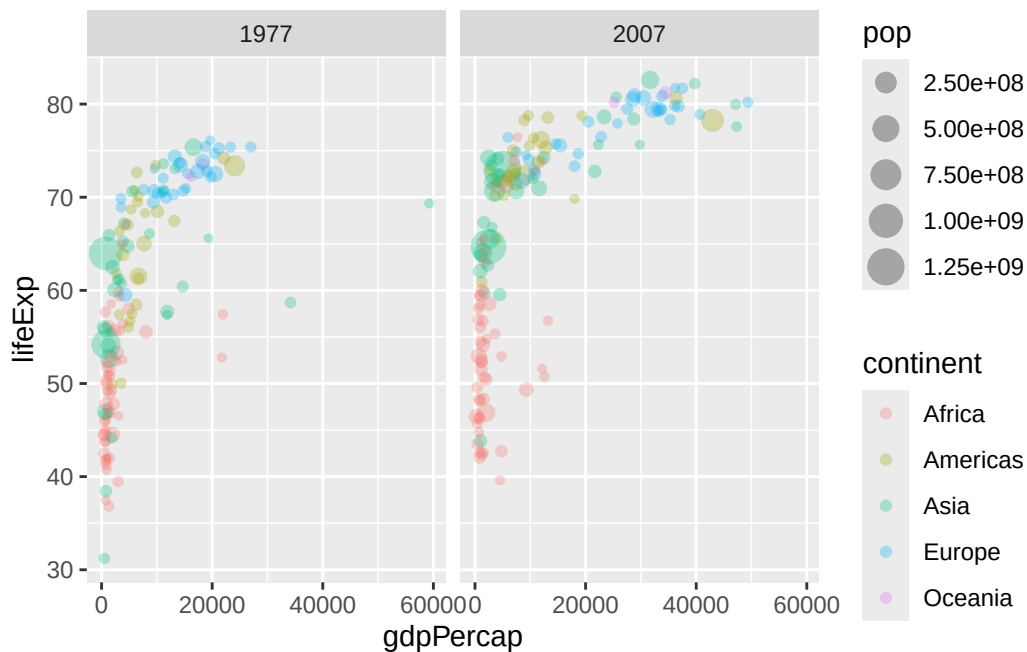
Make a plot for years 1974 and 1978 only (not all the years in the dataset).

Q. First use the **dplyr** package and the **filter()** function from that package to extract the rows from the year 1978.

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year==1977)
g <- filter(gapminder, year == 2007 | year == 1977)
```

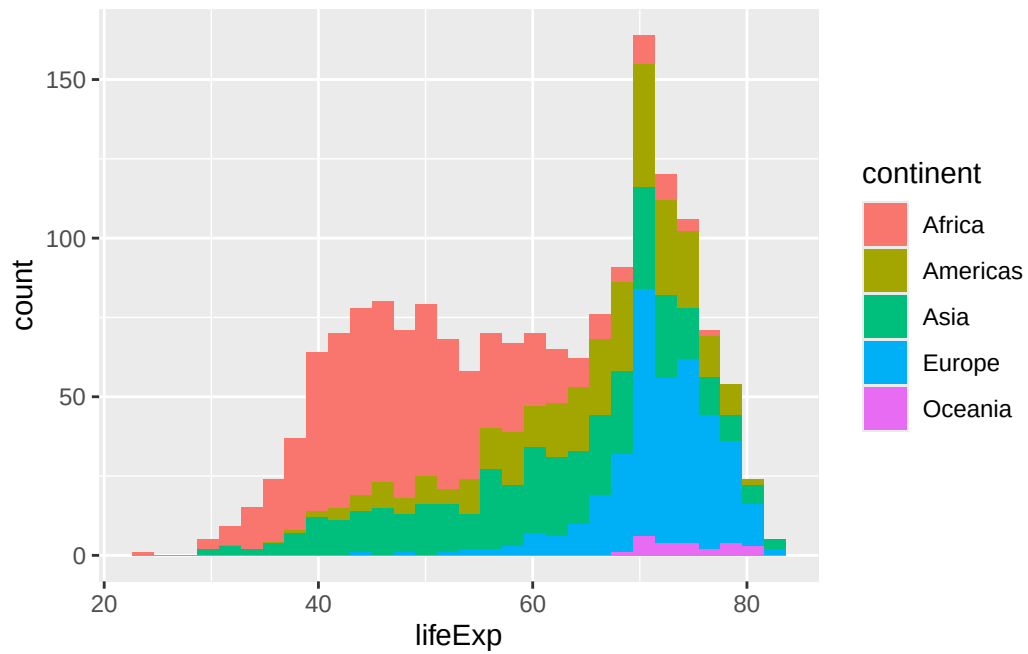
```
ggplot(g) +
  aes(x = gdpPerCap, y = lifeExp, col = continent, size = pop) +
  geom_point(alpha=0.3) +
  facet_wrap(~year)
```



Q. Make a histogram of lifeExp colored by continent (try using **fill=continent** or **col=continent**)

```
ggplot(gapminder) +
  aes(lifeExp) +
  geom_histogram(aes(fill=continent))
```

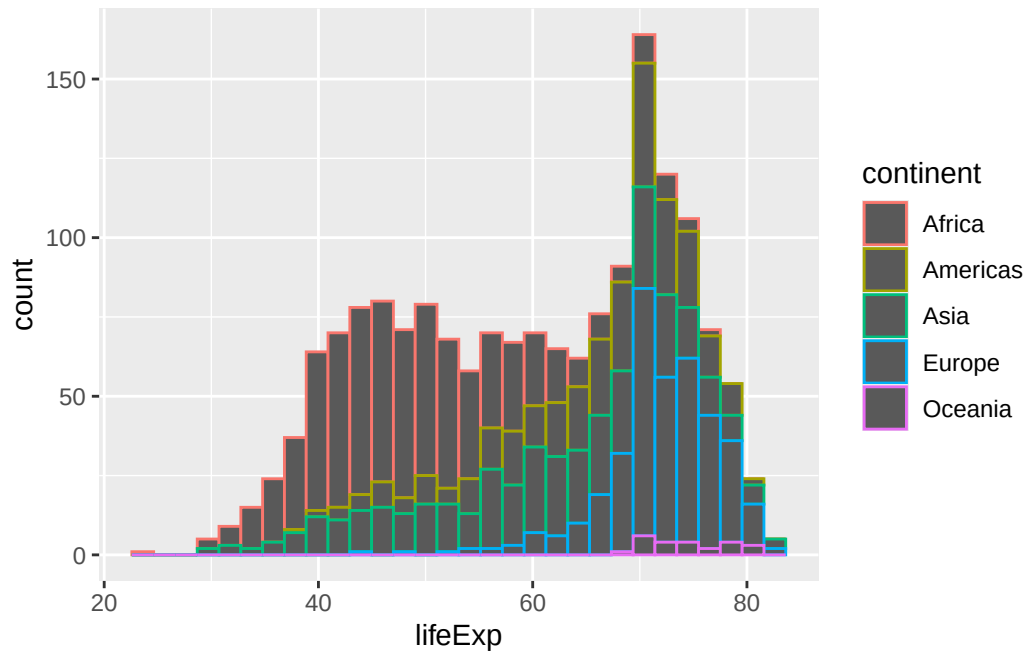
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



OR

```
ggplot(gapminder) +  
  aes(lifeExp) +  
  geom_histogram(aes(col=continent))
```

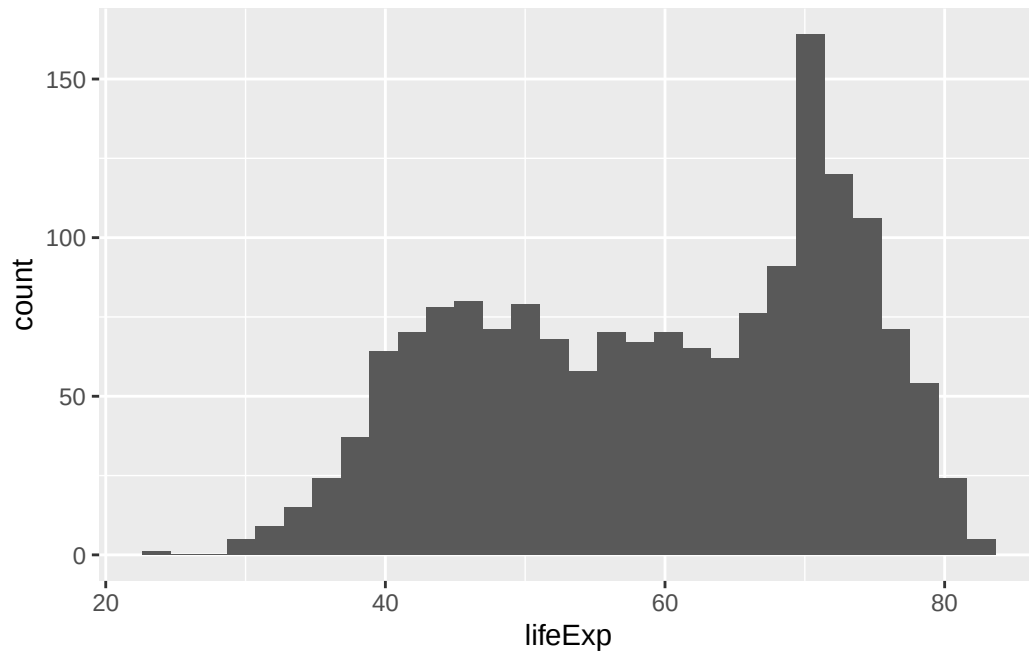
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +  
  aes(lifeExp) +  
  geom_histogram()
```

``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
facet_wrap(~continent)
```

```
<ggproto object: Class FacetWrap, Facet, gg>
  attach_axes: function
  attach_strips: function
  compute_layout: function
  draw_back: function
  draw_front: function
  draw_labels: function
  draw_panel_content: function
  draw_panels: function
  finish_data: function
  format_strip_labels: function
  init_gtable: function
  init_scales: function
  map_data: function
  params: list
  set_panel_size: function
  setup_data: function
  setup_panel_params: function
  setup_params: function
  shrink: TRUE
```

```
train_scales: function
vars: function
super: <ggproto object: Class FacetWrap, Facet, gg>
```